

# NEWTON BASKARAN

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## SUMMARY

Detail-oriented Mechanical Engineer with a strong background in Product Design, Manufacturing, and Process Improvement. Experienced in optimizing processes, improving efficiency, and maximizing productivity within various operations. Eager to contribute effectively to an organization while continuously upskilling to adapt to the ever-changing industrial landscape.

## EDUCATION

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| <b>M.S, Mechanical Engineering;</b> Arizona State University, Tempe, AZ                                                                                                                           | <b>Aug 2022 – May 2024</b> |
| Relevant Coursework: Modern Manufacturing, Advanced Energy System Design, Stress Analysis, Battery Materials Science & Engineering, Regression Analysis, and Analysis & Modelling of Fluid Flows. | 3.43 GPA                   |
| <b>B.E, Mechanical Engineering;</b> Anna University, Chennai, India                                                                                                                               | <b>Aug 2017 – May 2021</b> |
| Relevant Coursework: Lean Manufacturing, Material Science, Manufacturing Technology, Kinematics of Machinery, Engineering Mechanics, Strength of Materials, and Engineering Metallurgy.           | 3.20 GPA                   |

## TECHNICAL SKILLS

**CAD/CAM/CAE Tools:** AutoCAD, SolidWorks, Creo, NX, Vis Mockup, Rhino, Ansys, Autodesk Fusion, Autodesk Inventor.

**Programming Languages & Expertise:** Python, MATLAB, SQL, C++, Casting, Lean Six Sigma, QA, GD&T, BOM & PPAP Management.

## PROFESSIONAL EXPERIENCE

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| <b>Caterpillar, Tucson, AZ; Manufacturing Engineer (Contracted through Midsys Inc.)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Oct 2024 – Present</b>   |
| <ul style="list-style-type: none"><li>Developed and standardized processes to improve the quality of supplier parts like cores, chills, and basins supporting the foundry process, while managing the design change levels and maintaining the MBOM.</li><li>Lead virtual manufacturing for ultra-class trucks, working with design teams across systems conducting design audits.</li><li>Lead commonality study across the large mining truck lineup to identify and design modules usable across machines, enabling significant cost savings.</li></ul>                                                                                                             |                             |
| <b>Arizona State University, Tempe, AZ; Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>June 2024 – Oct 2024</b> |
| <ul style="list-style-type: none"><li>Developed a portable ultrasonic nebulizer, from the ground up, managing a multidisciplinary team of 6 engineers across design, prototyping, and manufacturing phases, ensuring product compliance with health and safety standards.</li><li>Lead the design team to develop product layouts and working prototypes aimed at scalable mass production.</li><li>Built relationships with outsourcing companies to source essential components, while creating and managing the Bill of Materials (BOM) to ensure smooth operations.</li></ul>                                                                                      |                             |
| <b>Infinite Solutions, Chennai, India; Mechanical Design Engineer &amp; Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Mar 2020 - Jul 2022</b>  |
| <b>Mechanical Design Intern (Mar 2020 – May 2021), Mechanical Design Engineer (Jun 2021 – Jul 2022)</b> <ul style="list-style-type: none"><li>Leveraged mechanical design expertise on projects for companies like Nissan, TVS, Daimler, and Ford, enhancing product performance and reliability.</li><li>Collaborated with senior Design Engineers on FEA/CFD projects for durability assessments, thermal behavior, and sheet metal stamping simulations.</li><li>Developed comprehensive process workflows integrating Lean and Six Sigma methodologies such as DMAIC to meet customer specifications and developed Standard Operating Procedures (SOPs).</li></ul> |                             |

## ACADEMIC PROJECTS

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| <b>Aerodynamic Analysis of NACA 4412 Airfoil Using CFD and NASA Data Comparison</b>                                                                                                                                                                                                                                                                                                                                          | <b>Feb 2024 – Apr 2024</b>  |
| <ul style="list-style-type: none"><li>Matched NASA data with lift and drag coefficients of 1.694 vs. 1.721 and 0.0278 vs. 0.0286 at 13° angle of attack, while analyzing stall onset and drag trends from 0° to 13°.</li><li>Simulated boundary layer effects and flow separation using the Spalart-Allmaras turbulence model in ANSYS Fluent, achieving results closely aligned with experimental data.</li></ul>           |                             |
| <b>Optimizing Parallel Plate-Fin Heat Sinks via Simulation and Machine Learning Techniques</b>                                                                                                                                                                                                                                                                                                                               | <b>Sept 2023 – Nov 2023</b> |
| <ul style="list-style-type: none"><li>Designed ANSYS simulations to evaluate heat transfer coefficients and pressure drops across varied Reynolds numbers and fin geometries, ensuring robust thermal analysis.</li><li>Developed MATLAB-based ANN models (<math>R &gt; 0.9</math>) and CNN models for image-based geometry recognition, optimizing heat sink configurations for performance.</li></ul>                      |                             |
| <b>Fabricated a go-kart for Bharath Formula Karting</b>                                                                                                                                                                                                                                                                                                                                                                      | <b>Sept 2020 – Jan 2021</b> |
| <ul style="list-style-type: none"><li>Engineered and manufactured a high-performance 155cc go-kart as part of a 20-member team, ensuring compliance with competition standards and achieving optimal racing conditions for the National Formula Karting event.</li><li>Led the business management team, developing and presenting a strategic industrial business plan, winning the Best Business Proposal Award.</li></ul> |                             |

## CERTIFICATES AND COURSES

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|------------------------------------------------------------------------|-----------------|
| • Lean Six Sigma Green Belt: Awareness and its Implementation, IMTE    | <b>Feb 2022</b> |
| • Thermodynamics and Phase Equilibria, Stanford, edX                   | <b>Jan 2022</b> |
| • Product Design, Prototyping and Testing, University of Maryland, edX | <b>Dec 2021</b> |